The Java™ Tutorials

The Java Tutorials have been written for JDK 8. Examples and practices described in this page don't take advantage of improvements introduced in later releases and might use technology no longer available.
See Dev.java for updated tutorials taking advantage of the latest releases.
See Java Language Changes for a summary of updated language features in Java SE 9 and subsequent releases.
See JDK Release Notes for information about new features, enhancements, and removed or deprecated options for all JDK releases.

Trail: Creating a GUI With Swing

Also known as *The Swing Tutorial*

This trail tells you how to create graphical user interfaces (GUIs) for applications and applets, using the Swing components. If you would like to incorporate JavaFX into your Swing application, please see Integrating JavaFX into Swing Applications.

Getting Started with Swing is a quick start lesson. First it gives you a bit of background about Swing. Then it tells you how to compile and run programs that use Swing components.

Learning Swing with the NetBeans IDE is the fastest and easiest way to begin working with Swing. This lesson explores the NetBeans IDE's GUI builder, a powerful feature that lets you visually construct your Graphical User Interfaces.

Using Swing Components tells you how to use each of the Swing components — buttons, tables, text components, and all the rest. It also tells you how to use borders and icons.

Concurrency in Swing discusses concurrency as it applies to Swing programming. Information on the event dispatch thread and the SwingWorker class are included.

Using Other Swing Features tells you how to use actions, timers, and the system tray; how to integrate with the desktop class, how to support assistive technologies, how to print tables and text, how to create a splash screen, and how to use modality in dialogs.

Laying Out Components Within a Container tells you how to choose a layout manager, how to use each of the layout manager classes the Java platform provides, how to use absolute positioning instead of a layout manager, and how to create your own layout manager.

Modifying the Look and Feel tells you how to specify the look and feel of Swing components.

Drag and Drop and Data Transfer tells you what you need to know to implement data transfer in your application.

Writing Event Listeners tells you how to handle events in your programs.

Performing Custom Painting gives you information on painting your own Swing components. It discusses painting issues specific to Swing components, provides an overview of painting concepts, and has examples of custom components that paint themselves.

Other UI-Related Trails

Although this is the main trail for learning about GUIs, it isn't the only trail with UI-related information.

- 2D Graphics, which describes the 2D graphics features available in the JDK.
- Sound, which discusses the sound capabilities available in the JDK.
- Java Applets, which describes API available only to applets.
- Essential Java Classes, which covers many topics, including properties and the standard I/O streams.
- The JavaFX Documentation, which describes how to build UIs with JavaFX.
- The Bonus trail contains Full-Screen Exclusive Mode API, a lesson that describes how to use API introduced in v1.4 to render graphics directly to the screen.

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